Industrial Internship Report on

" URL Shortener System using Python"

Prepared by

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time. My project was (Tell about ur Project) This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

TABLE OF CONTENTS

1 Preface

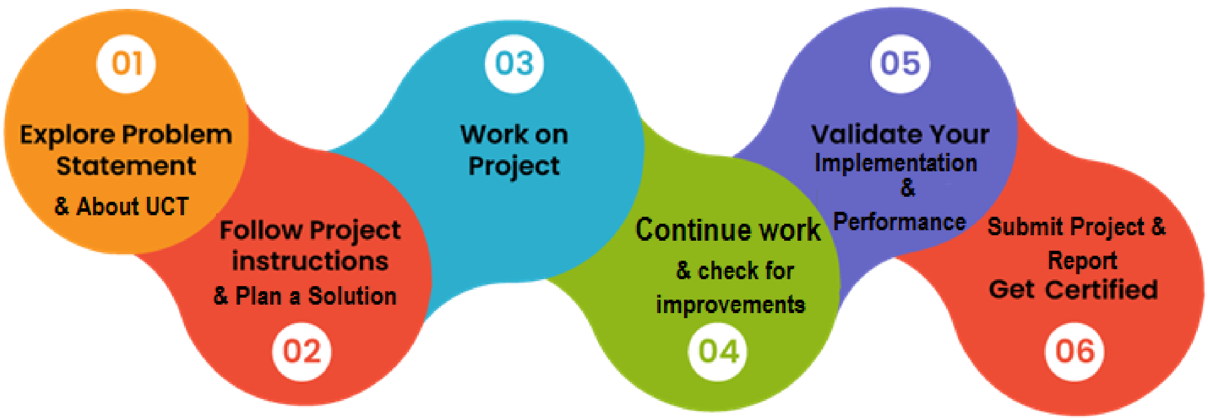
Summary of the whole 6 weeks' work.

About need of relevant Internship in career development.

Brief about Your project/problem statement.

Opportunity given by USC/UCT.

How Program was planned



Your Learnings and overall experience.

Thank to all (with names), who have helped you directly or indirectly.

Your message to your juniors and peers.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

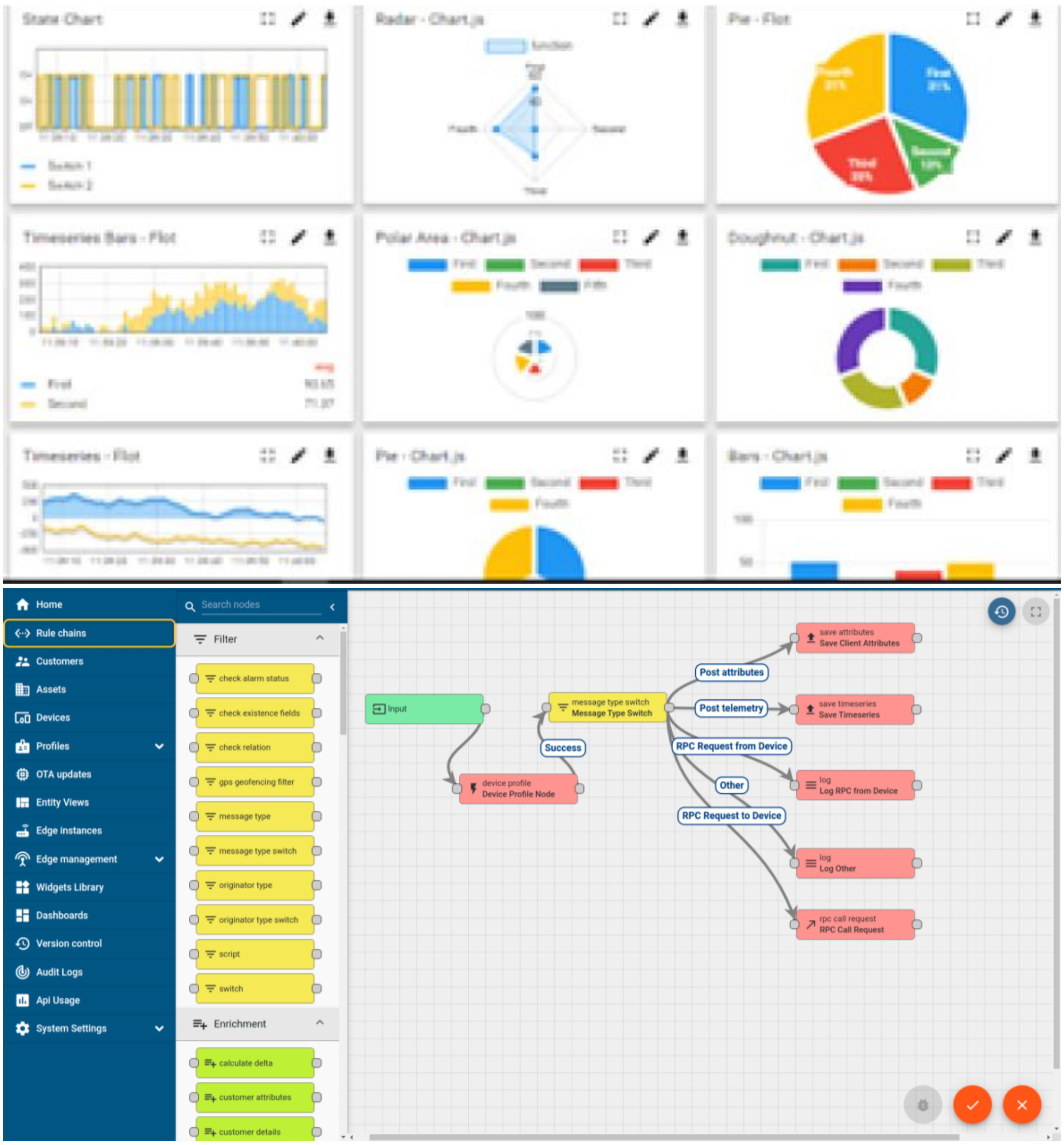
i. UCT IoT Platform ( )

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

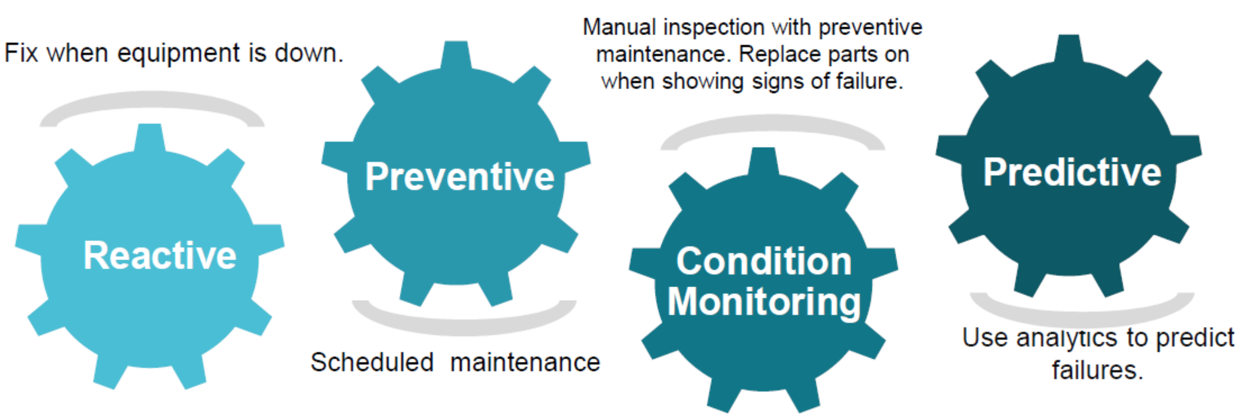
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

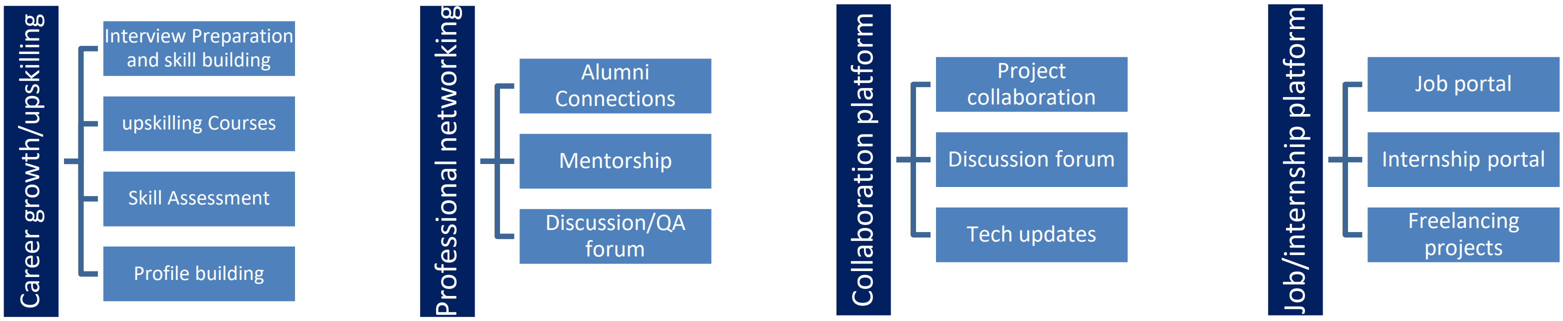
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

☛ get practical experience of working in the industry.

☛ to solve real world problems.

☛ to have improved job prospects.

☛ to have Improved understanding of our field and its applications.

☛ to have Personal growth like better communication and problem solving.

2.5 Glossary

Terms	Acronym

3 Problem Statement

The problem statement assigned during the internship was:

"To design and develop a system that converts long URLs into short URLs and redirects users efficiently."

3.1.1 Challenges:

- Generating unique short links
- Avoiding duplication
- Fast redirection
- Managing URL database

In today's digital world, URLs (Uniform Resource Locators) are often very long, complex, and difficult to share or remember. These lengthy URLs are not user-friendly, especially when sharing links through platforms like social media, emails, or messages. Long URLs can also look unprofessional and may discourage users from clicking them.

The problem is to design and develop a system that can **convert long URLs into short, simple, and unique links**, while ensuring that users are still redirected to the original website correctly and efficiently.

The system must address the following challenges:

- Generate a **unique short URL** for each long URL
- Avoid duplication or conflicts between generated links
- Ensure **fast and accurate redirection**
- Store and manage URL mappings efficiently using a database
- Handle invalid or incorrect inputs properly

The goal is to create a **reliable, efficient, and user-friendly URL shortening system** that simplifies link sharing while maintaining performance and accuracy.

4 Existing and Proposed solution

4.1 4.1 Existing Solutions

There are several URL shortening services available such as **Bitly, TinyURL, and Rebrandly**. These platforms allow users to convert long URLs into short and shareable links.

4.1.1 Limitations of Existing Solutions:

- Many advanced features (analytics, customization) are **paid**
- Limited control over link management
- Dependency on third-party platforms
- Some services show advertisements
- Privacy and data security concerns
- Limited customization of short URLs

4.2 4.2 Proposed Solution

The proposed system is a **Python-based URL Shortener** that converts long URLs into short, unique links and redirects users to the original URL when accessed.

4.2.1 Features of Proposed System:

- Generates **unique short URLs** using algorithms
- Stores URL mappings in a **database**
- Provides **fast and accurate redirection**
- Simple and user-friendly interface
- Handles invalid inputs properly

4.3 4.3 Value Addition

The proposed solution improves over existing systems in the following ways:

- **Cost-effective** – No subscription required
- **Customizable** – Can be modified as per user needs
- **Lightweight system** – Uses minimal resources
- **Improved control** – Full access to data and logic
- **Secure** – Can implement basic security features
- **Scalable** – Can be extended with features like analytics and login system

4.4 Code submission (Github link)

https://github.com/bhagyshriahirrao5-commits/upskillCampus

4.5 Report submission (Github link) : first make placeholder, copy the link.

https://github.com/bhagyshriahirrao5-commits/upskillCampus

5 Proposed Design/ Model

The proposed system is a **Python-based URL Shortener** designed to convert long URLs into short and manageable links. The system follows a structured workflow consisting of input, processing, storage, and output stages.

5.1 Design Flow of the System

The working of the system can be divided into the following stages:

5.1.1 1. Input Stage

- The user enters a **long URL** into the system through a simple user interface.
- The system validates whether the entered URL is correct and properly formatted.

5.1.2 2. Processing Stage

- After validation, the system generates a **unique short code** for the given URL.
- This short code can be created using techniques such as:
  - Random string generation
  - Hashing algorithms
- The system ensures that the generated code is **unique** and not already used.

5.1.3 3. Storage Stage

- The original URL and its corresponding short code are stored in a **database**.
- This mapping helps in retrieving the original URL later.

5.1.4 4. Output Stage

- The system provides a **shortened URL** to the user.
- Example:
  - Original URL: https://example.com/very-long-link
  - Short URL: https://short.ly/abc123

5.1.5 5. Redirection Stage

- When the user clicks the short URL:
  - The system searches the database
  - Retrieves the original URL
  - Redirects the user to the actual website

5.2 High Level Diagram (if applicable)

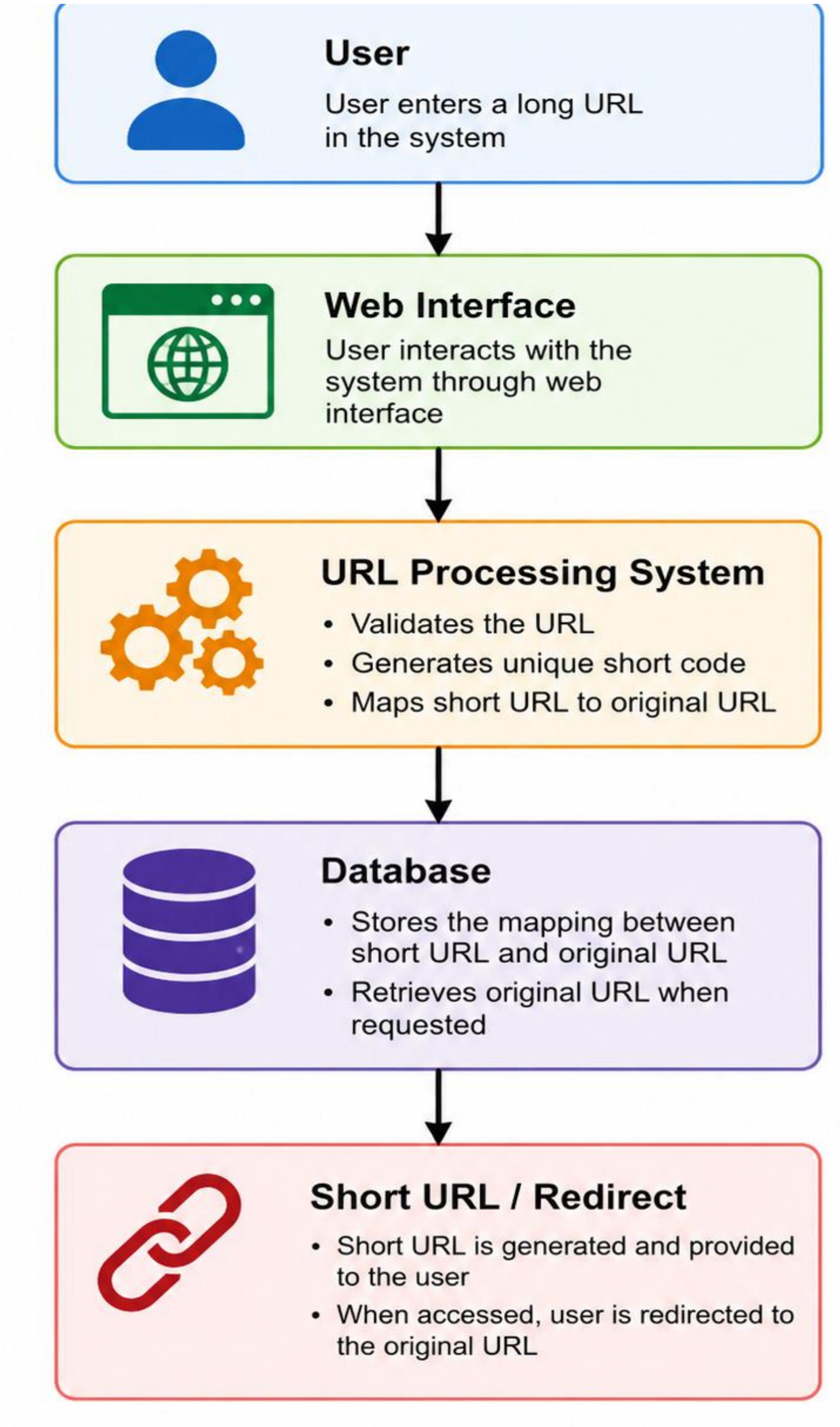


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.3 Low Level Diagram (if applicable)

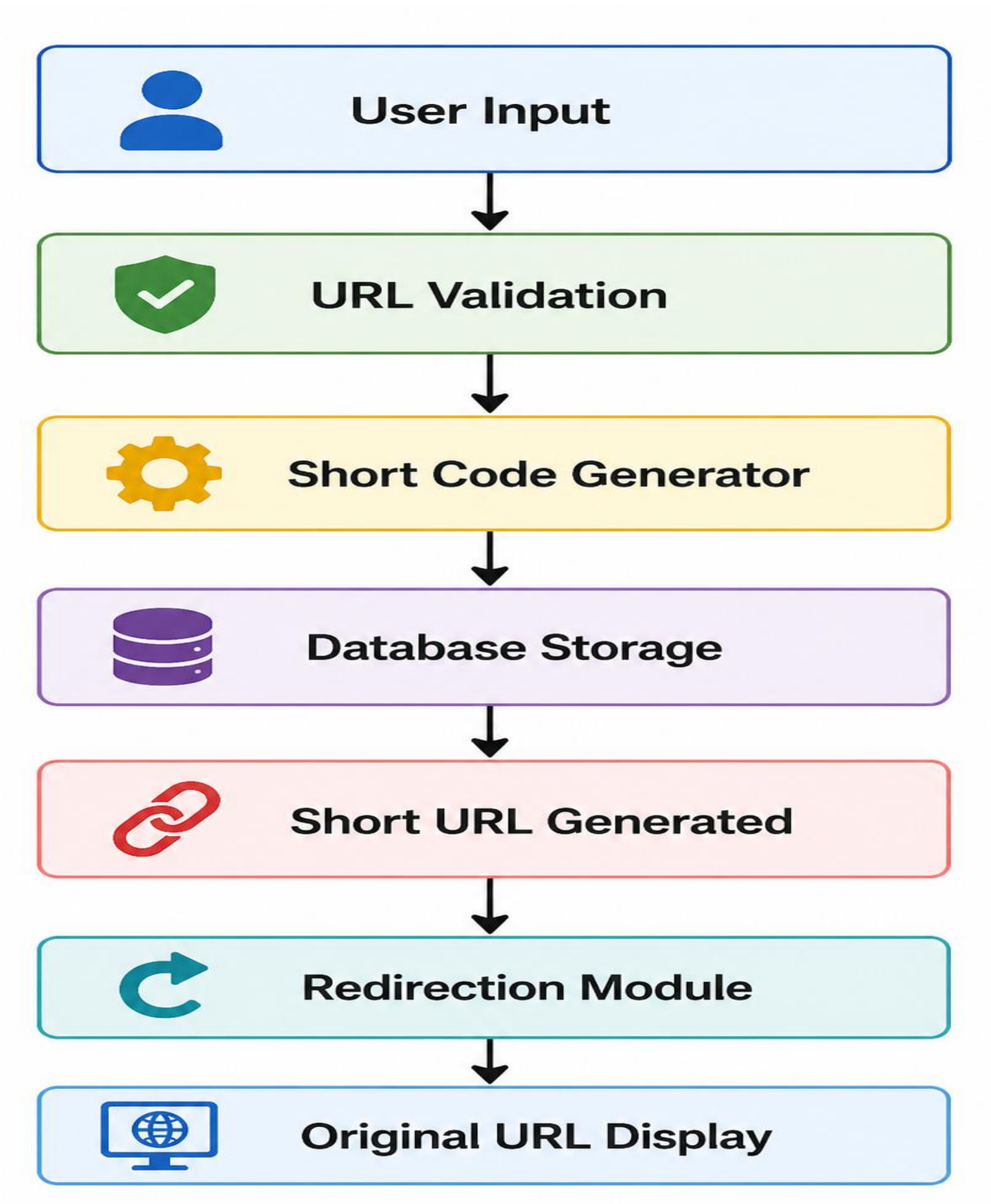


Figure 2: LOW LEVEL DIAGRAM OF THE SYSTEM

5.4 Interfaces (if applicable)

The URL Shortener system includes different interfaces and internal modules that communicate with each other to perform the required operations efficiently.

1. User Interface (UI)

- A simple interface where the user enters a **long URL**
- Displays the **generated short URL**
- Can be implemented using:
  - Web interface (HTML/CSS)

2. Input/Output Interface

- **Input:** Long URL entered by the user
- **Output:** Shortened URL and redirection to original URL

3. System Modules (Block Diagram Explanation)

The system consists of the following modules:

- **URL Validation Module:**
  Checks whether the entered URL is valid or not
- **Short Code Generator:**
  Generates a unique short code for each URL
- **Database Module:**
  Stores mapping between original URL and short URL
- **Redirection Module:**
  Redirects the user to the original URL when short URL is accessed

4. Data Flow (Flow Explanation)

1. User enters a long URL
2. URL is validated by the system
3. Unique short code is generated
4. URL mapping is stored in database
5. Short URL is generated and displayed
6. On accessing short URL → system redirects to original URL

5. Protocols Used

- **HTTP/HTTPS:**
  Used for communication between user and system
- **Database Queries (SQL):**
  Used for storing and retrieving URL data

6 Performance Test

Performance testing is an important part of the system as it determines whether the project can be used in real-world industrial applications. The URL Shortener system was evaluated based on constraints like speed, memory usage, accuracy, and scalability.

6.1 Constraints Identified

- **Speed (Response Time):** Fast generation of short URLs and quick redirection
- **Memory Usage:** Efficient storage of large numbers of URLs
- **Accuracy:** Correct mapping between short and original URLs
- **Scalability:** Ability to handle multiple users and requests
- **Database Performance:** Fast retrieval and storage operations

6.2 Test Plan/ Test Cases

Test Case ID	Test Description	Input Example	Expected Output	Result
TC01	Valid URL input	https://example.com	Short URL generated	Pass
TC02	Invalid URL input	abc123	Error message	Pass
TC03	Duplicate URL input	Same URL entered twice	Same/unique short URL generated	Pass
TC04	Redirection test	Click short URL	Redirect to original URL	Pass
TC05	Empty input	(blank)	Validation error	Pass

6.3 Test Procedure

- Enter a valid long URL into the system
- Verify that a short URL is generated
- Click the generated short URL
- Check if it redirects to the correct original URL
- Repeat with invalid and duplicate inputs
- Observe system response time and behavior

6.4 Performance Outcome

- The system generates short URLs in **very less time (milliseconds)**

- Redirection is **fast and accurate**
- No errors observed for valid inputs
- Proper error handling for invalid inputs
- System performs efficiently for small to medium datasets

7 Limitations

Due to limited resources, large-scale testing (high traffic, thousands of users) was not performed. However, the system can be improved for scalability using advanced technologies.

8 My learnings

During this internship, I gained valuable practical knowledge and hands-on experience in developing a real-world project. Working on the **URL Shortener System using Python** helped me understand how theoretical concepts are applied in real applications.

I improved my technical skills in **Python programming, database management, and backend logic development**. I also learned how to design a system, handle user input, and implement efficient algorithms for generating unique short URLs.

Apart from technical skills, this internship enhanced my **problem-solving ability, logical thinking, and debugging skills**. I also learned the importance of writing clean code and testing the system properly.

I developed better **time management and self-learning skills** while completing the project within the given timeline.

Overall, this internship has increased my confidence and prepared me for future opportunities in the software industry. It has helped me understand industry expectations and will be beneficial for my career growth.

9 Future work scope

The current URL Shortener system provides basic functionality for converting long URLs into short links and redirecting users efficiently. However, there are several enhancements that can be implemented in the future to improve the system.

Some possible future improvements include:

- **User Authentication System:**
  Adding login and signup features so users can manage their own shortened URLs.
- **Custom Short URLs:**
  Allowing users to create personalized short links instead of randomly generated ones.
- **Analytics and Tracking:**
  Implementing features to track the number of clicks, user location, and usage statistics of each short URL.
- **Cloud Deployment:**
  Hosting the system on cloud platforms to improve scalability and availability.
- **Security Enhancements:**
  Adding encryption and protection mechanisms to prevent misuse and malicious links.
- **QR Code Generation:**
  Generating QR codes for shortened URLs for easy sharing.
- **Improved Database Management:**
  Using advanced databases for handling large-scale data efficiently.